

Bitwise differential trail through **Blink-128** found by SAT

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1 Blink-128

Maximal differential probability: 2^{-32}

0	0	0	0	0	0	0	8	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	8			
0	0	0	0	0	0	0	8	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	8			
rk																															
0	0	0	0	0	0	0	8	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	8			
0	0	0	0	0	0	0	8	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	8			
padRound																															
0	0	0	0	0	2	0	2	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0			
0	0	0	0	0	2	0	2	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0			
padRound																															
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0			
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0			
padRound																															
2	0	0	2	2	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
2	0	0	2	2	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
SubCells																															
8	0	0	2	8	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
8	0	0	2	8	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
MixColumns																															
0	0	0	0	0	2	0	0	8	0	0	2	8	0	0	0	8	0	0	2	8	2	0	0	8	0	0	2	8	2	0	0
0	0	0	0	0	2	0	0	8	0	0	2	8	0	0	0	8	0	0	2	8	2	0	0	8	0	0	2	8	2	0	0
rk																															
0	0	0	0	0	2	0	0	8	0	0	2	8	0	0	0	8	0	0	2	8	2	0	0	8	0	0	2	8	2	0	0
0	0	0	0	0	2	0	0	8	0	0	2	8	0	0	0	8	0	0	2	8	2	0	0	8	0	0	2	8	2	0	0
Permutation																															
2	0	8	8	0	0	8	0	2	8	0	0	8	0	0	2	0	0	0	8	0	0	0	0	0	2	0	0	0	2	2	0
2	0	8	8	0	0	8	0	2	8	0	0	8	0	0	2	0	0	0	8	0	0	0	0	0	2	0	0	0	2	2	0

2 FwdRound

Maximal differential probability: $2^{-8.0}$

0	0	0	0	0	0	0	8	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	8	
SubCells																														
0	0	0	0	0	0	0	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2
MixColumns																														
0	0	0	0	0	0	0	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2
RK																														
0	0	0	0	0	0	0	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2
RK																														
0	0	0	0	0	0	0	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2
Permutation																														
0	0	0	0	0	2	0	2	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	2	0	2	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

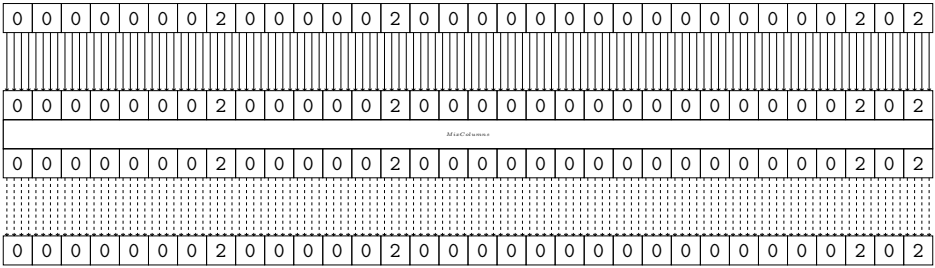
3 SubCells

Maximal differential probability: $2^{-8.0}$

0	0	0	0	0	0	0	8	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	8	
0	0	0	0	0	0	0	8	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	8
∂B_{102}	∂B_{103}	∂B_{104}	∂B_{105}	∂B_{106}	∂B_{107}	∂B_{108}	∂B_{109}	∂B_{110}	∂B_{111}	∂B_{112}	∂B_{113}	∂B_{114}	∂B_{115}	∂B_{116}	∂B_{117}	∂B_{118}	∂B_{119}	∂B_{120}	∂B_{121}	∂B_{122}	∂B_{123}	∂B_{124}	∂B_{125}	∂B_{126}	∂B_{127}	∂B_{128}	∂B_{129}		
0	0	0	0	0	0	0	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2
0	0	0	0	0	0	0	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2

4 MixColumns

Maximal differential probability: $2^{-0.0}$



5 FwdRound

Maximal differential probability: $2^{-8.0}$

[illegible]

6 SubCells

Maximal differential probability: $2^{-8.0}$ [illegible]

7 MixColumns

Maximal differential probability: $2^{-0.0}$ [illegible]

8 FwdRound

Maximal differential probability: $2^{-8.0}$

2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	2	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0
<i>BulkCells</i>																						
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
<i>MiscColumns</i>																						
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
<i>OK</i>																						
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
<i>OK</i>																						
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
2	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
<i>Permutation</i>																						
2	0	0	2	2	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0
2	0	0	2	2	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0

9 SubCells

Maximal differential probability: $2^{-8.0}$ [illegible]

10 MixColumns

Maximal differential probability: $2^{-0.0}$ [illegible]

11 SubCells

Maximal differential probability: $2^{-8.0}$ [illegible]

12 MixColumns

Maximal differential probability: $2^{-0.0}$

8	0	0	2	8	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
8	0	0	2	8	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
MaxColumns																															
0	0	0	0	0	2	0	0	8	0	0	2	8	0	0	0	8	0	0	2	8	2	0	0	8	0	0	2	8	2	0	0
0	0	0	0	0	2	0	0	8	0	0	2	8	0	0	0	8	0	0	2	8	2	0	0	8	0	0	2	8	2	0	0